

CallShield AI — Technical System Documentation

Architecture Overview, Technology Stack, Folder Structure & Log Storage Reference

1. Official Project Deployment & Repository Links

Resource	URL / Access Path	Status
Live Web Application	https://scam-audio-full-project.vercel.app	Active / Live
GitHub Repository	https://github.com/manav3009/scam-audio-detection	Public Repo
Android Package (.apk)	CallShield_AI_v1.0.apk (Desktop & Downloads)	Standalone APK
Web APK Download Link	https://scam-audio-full-project.vercel.app/static/downloads/CallShield_AI_v1.0.apk	Direct Web DL

2. Technology Stack & Framework Specifications

Backend Framework	Python 3.10+ & Flask 2.3	Handles REST API routes, session security, and detector orchestration.
Frontend UI / UX	HTML5, CSS3, JavaScript	Responsive Glassmorphism interface optimized for Phone, Tablet, & PC.
Speech-to-Text Engine	Web Speech API	Browser-native real-time speech recognition for live calls & mic audio.
Audio Processing	Web Audio API & MediaRecorder	Generates real-time waveform visualizers and binary audio blob recording.
Scam Detection AI Engine	Heuristic NLP Engine	Multi-tier pattern scanner scoring urgency, banking, and OTP risk (0-100%).
AI Cybersecurity Chatbot	DuckDuckGo API Engine	Real-time cybersecurity search engine fetching up-to-date threat intel.
Android Native Layer	Java & Android SDK (API 21-34)	Full-screen WebView container, native mic permissions & call state listener.
Cloud Infrastructure	Vercel Serverless Hosting	Production deployment with automated Git integration & global CDN.
Build & Packaging	Gradle 8.13	Builds signed standalone Android debug/release APK binaries.

3. Directory & Folder Structure

A. Web Application Module Breakdown (Scam_audio_Full_Project)

Path / File Name	Module Category	Description & Functional Purpose
app.py	App Core	Main Flask application factory, secret key config, and blueprint registration.
api/index.py	Vercel Entry	Serverless entry point for cloud deployment on Vercel platform.
core/detector.py	AI Engine	Scam NLP analyzer calculating risk scores, red flags, & warning messages.
core/chatbot_data.py	AI Engine	Cybersecurity knowledge base parser and web search integration.
core/test_data.py	Data Layer	Sample scam audio dataset & registry for newly recorded live calls.
routes/views_routes.py	API / View	Renders index, dashboard, login, and main application pages.

routes/analysis_routes.py	API / View	Handles live mic streaming, recorded audio analysis, and file uploads.
routes/chatbot_routes.py	API / View	Processes chat queries and returns AI security responses.
routes/auth_routes.py	Security	Manages Admin and Student role-based login sessions.
templates/base.html	UI Template	Master HTML layout containing responsive topbar, drawer, & footer.
templates/index.html	UI Template	Main landing page featuring hero section, feature cards, & quick scanner.
templates/login.html	UI Template	Authentication portal for Student and Admin users.
templates/dashboard.html	UI Template	Student & Admin statistics dashboard with scam analytics.
templates/modules/live_call.html	UI Module	Real-time live call scanner with speaker split & wave animation.
templates/modules/recorded.html	UI Module	Offline audio file uploader and sample scam dropdown selector.
templates/modules/chatbot.html	UI Module	Interactive cybersecurity AI chat interface with history storage.
templates/modules/awareness.html	UI Module	Scam awareness education guide detailing common fraud techniques.
static/css/ & js/	Static Assets	Custom stylesheets and client-side JavaScript for Web Speech API.
vercel.json & requirements.txt	Config	Vercel route configuration and Python dependencies manifest.

B. Android Application Module Breakdown (CallShield_Android_App)

app/build.gradle	Build Config	SDK configuration (minSdk 21 / targetSdk 34) & AndroidX dependencies.
AndroidManifest.xml	Permissions	Declares RECORD_AUDIO, INTERNET, and CALL_STATE hardware access.
MainActivity.java	Native Code	Hosts full-screen Android WebView, WebChromeClient mic bridge, & telephony listener.
build.gradle & properties	Build Config	Project build rules and Gradle 8.13 distribution configuration.

4. Logs & Detection History Storage Guide

Log / History Type	Storage Mechanism & Location	Persistence & Retention Policy
Chatbot Conversation History	Browser localStorage (Key: callshield_chat_history)	Saved client-side on device. Persists across browser refreshes and restarts.
Live Call Transcripts	In-Memory Registry (TEST_AUDIO_FILES in test_data.py)	Automatically populates the Recorded Audio dropdown list during active sessions.
Vercel Cloud Server Logs	Vercel Functions Dashboard (Deployment: scam-audio-full-project)	Maintains production HTTP access logs, status codes, and server execution traces.
Android App Native Logs	Android Logcat (Filter Tag: CallShield_App)	Captures microphone permission grants, call state events, & WebView console output.